

ECE 315: Signals and Systems

Homework 6

Instructor: Dongsheng Ding

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This homework is due on **4/6, 11:59pm**. For analytical problems, write your solutions independently by hand. For coding problems, attach your source code and answers. Combine them in one PDF document and submit on CANVAS. Please ensure that your PDF is clearly readable on a computer.

1. (1pts: Periodicity of discrete-time signals) Consider the following discrete-time signals:

- $2 \sin(\frac{\pi}{3}n) + \cos(\frac{8\pi}{3}n)$
- $(\sin(\frac{\pi}{6}n))^2$

Determine whether they are periodic or aperiodic. For periodic signals, find their fundamental periods and write them in terms of harmonically related complex exponentials.

2. (2pts: Fourier series representation) Consider the following Fourier series coefficients for each discrete-time periodic signal $x[n]$ with period 6:

- $a_k = \cos(\frac{\pi}{3}k) + \sin(\frac{8\pi}{3}k)$
- $a_k = \begin{cases} \sin(\frac{\pi}{2}k) & \text{when } 0 \leq k \leq 4 \\ 0 & \text{when } k = 5 \end{cases}$

Determine the signal $x[n]$ for each case.

3. (2pts: Fourier series coefficients) Assume a discrete-time periodic signal $x[n]$ with period N has Fourier series representation,

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk \frac{2\pi}{N} n}.$$

We consider the following transformed signals:

- $y[n] = x[n] - x[n-1]$;
- $y[n] = x[n - \frac{N}{2}]$ (assume N is even);
- $y[n] = (-1)^n x[n]$ (assume N is even);
- $y[n] = \begin{cases} x[n] & n \text{ even} \\ 0 & n \text{ odd} \end{cases}$

Determine the Fourier series coefficients for each signal in terms of a_k .

4. (3pts: Fourier series properties) Consider the following discrete-time periodic signals with period 6:

- $x[n] = 1 + \sin(\frac{2\pi}{6}n)$;
- $y[n] = \sin(\frac{14\pi}{6}n + \frac{\pi}{6})$;
- $z[n] = x[n]y[n]$;

- Compute the Fourier series coefficients of $x[n]$;
- Compute the Fourier series coefficients of $y[n]$;
- Compute the Fourier series coefficients of $x[n]y[n]$, directly;
- Compute the Fourier series coefficients of $x[n]y[n]$ using the multiplication property, and determine whether it equals the one from (iii).

5. (2pts: Convolution sum, and Fourier series representation) Consider the following discrete-time LTI system with impulse response:

$$h[n] = \begin{cases} 1 & \text{when } n = 1 \\ -1 & \text{when } n = -1 \\ 0 & \text{otherwise} \end{cases}$$

Given that the input to the system is $x[n] = \sum_{k=-\infty}^{\infty} \delta[n-2k]$. We are interested in the output $y[n] = x[n] * h[n]$.

- Compute the output $y[n]$ using convolution sum;
 - Determine the Fourier series of the output $y[n]$.
6. Read Sections **9.1-2** and **9.7-1**, and plot the spectra of the periodic signals in **9.1-1** and **9.1-4** in **LSS**.